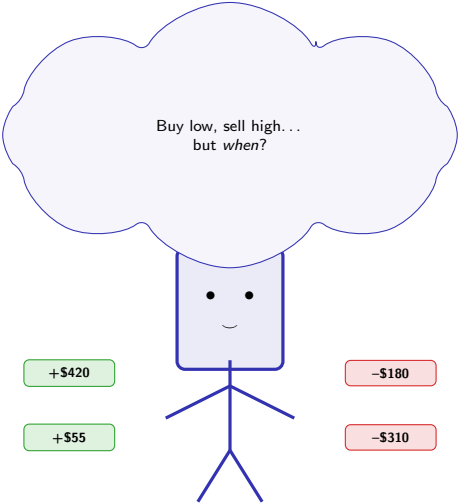
Reinforcement Learning

Learning to Act from Rewards

Methods & Algorithms

MSc Data Science – Spring 2026

Can a Machine Learn to Trade Stocks by Trial and Error?



RL agents learn from outcomes, not from labeled examples.

Learning Objectives

After this lecture, you will be able to:

1. **Formalize** sequential decisions as Markov Decision Processes
2. **Derive** the Bellman equation and implement Q-learning
3. **Evaluate** model-free RL methods (Q-learning vs SARSA vs PPO) for trading
4. **Critique** RL trading strategies accounting for transaction costs, non-stationarity, and offline constraints

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LOs at Bloom's levels 4–5: formalize, derive, analyze, apply.

What Is Reinforcement Learning?

An **agent** learns to make decisions by interacting with an **environment** and receiving **rewards**.

- Not supervised: no labeled "correct actions"
- Not unsupervised: there is a goal (maximize reward)
- Learning by **trial and error**

Key Terms

- **Agent**: the learner / decision-maker
- **Environment**: the world it acts in
- **State**: current situation
- **Action**: what the agent does
- **Reward**: feedback signal
- **Policy**: action selection strategy

RL sits between supervised and unsupervised learning: it has a goal but no labels.

Why Not Just Use Supervised Learning?

Supervised Learning

- Needs labeled data (x, y)
- "What is the correct action?" requires a label
- Assumes i.i.d. samples

Reinforcement Learning

- Learns from outcomes (reward signals)
- No one tells you the "right" trade
- Delayed rewards: today's trade affects tomorrow

In trading, you only know if a decision was good after the market moves.

What If There Is Only One State?

The Multi-Armed Bandit — simplest RL problem.

- k actions ("arms"), each with unknown reward
- Goal: maximize cumulative reward
- **Regret:** gap between optimal and actual reward
- No state transitions — just explore vs. exploit

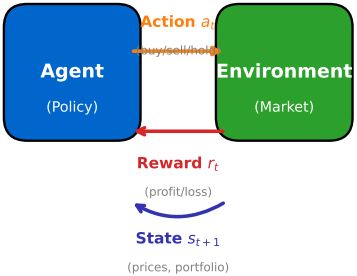
Finance Example

Which of k trading strategies should you allocate capital to? You only learn their quality by trying them.

Bandits are RL without state transitions. Add transitions → you get MDPs (next slide).

The Agent-Environment Loop

Reinforcement Learning: Agent-Environment Interaction



At each time step t:

Agent observes state, takes action, receives reward

https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L06_Embeddings_RL/03_rl_loop

State → Action → Reward → Next State → repeat.

Markov Decision Process (MDP)

Definition

An MDP is a tuple (S, A, P, R, γ):

- S: set of states $\quad A$: set of actions
- $P(s' \mid s, a)$: transition probability
- $R(s, a)$: reward function $\quad \gamma \in [0, 1]$: discount factor

Markov property: the future depends only on the current state, not the history.

$$P(S_{t+1} \mid S_t, A_t) = P(S_{t+1} \mid S_0, A_0, \ldots, S_t, A_t)$$

The MDP formalism is the mathematical foundation of all RL algorithms.

The Return and Value Functions

Return (discounted cumulative reward):

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

State-value function (how good is state s?):

$$V^\pi(s) = \mathbb{E}_\pi\left[G_t \mid S_t = s\right]$$

Action-value function (how good is action a in state s?):

$$Q^\pi(s, a) = \mathbb{E}_\pi\left[G_t \mid S_t = s,\ A_t = a\right]$$

γ close to 1: far-sighted agent. γ close to 0: myopic agent.

The Bellman Equation

The optimal action-value satisfies a **recursive** relationship:

$$Q^*(s,a) = R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \max_{a'} Q^*(s',a')$$

This expands the $\mathbb{E}[\cdot]$ form from the overview by substituting transition probabilities $P(s' \mid s, a)$.

Worked example ($\gamma = 0.9$, deterministic transitions):

- State A $\xrightarrow{\text{right}}$ State B, reward $= +1$
- State B $\xrightarrow{\text{right}}$ State C (terminal, $Q = 0$ by definition), reward $= +10$
- $Q(B, \text{right}) = 10 + 0.9 \times 0 = 10$
- $Q(A, \text{right}) = 1 + 0.9 \times 10 = 10$

The Bellman equation decomposes value into immediate reward + discounted future.

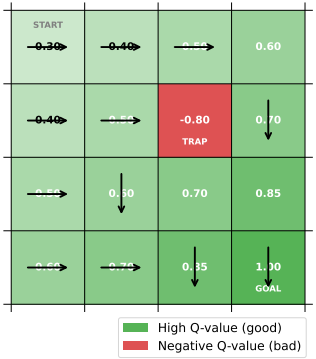
Q-Learning: Model-Free Temporal Difference

Update rule:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\left[R + \gamma \max_{a'} Q(s',a') - Q(s,a)\right]$$

- α: learning rate
- No transition model P needed!
- Converges to Q^* with sufficient exploration

Q-Learning: Grid World with Learned Q-Values



https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L06_Embeddings_RL/04_q_learning_grid

Q-learning is off-policy: it learns the optimal Q regardless of the behavior policy.

Exploration vs Exploitation

The Dilemma

- **Exploit**: choose the best known action
- **Explore**: try a random action to discover more
- Too much exploitation $\rightarrow$ stuck at local optimum
- Too much exploration $\rightarrow$ never converges

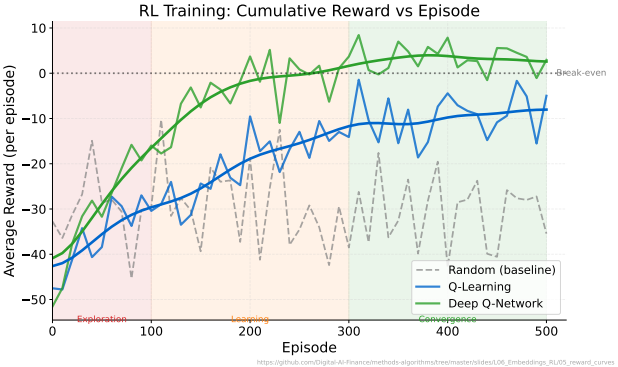
Epsilon-Greedy Policy

- With probability ϵ: random action
- With probability $1 - \epsilon$: greedy action
- **Decaying** ϵ: explore early, exploit later

Balancing exploration and exploitation is a fundamental challenge in RL.

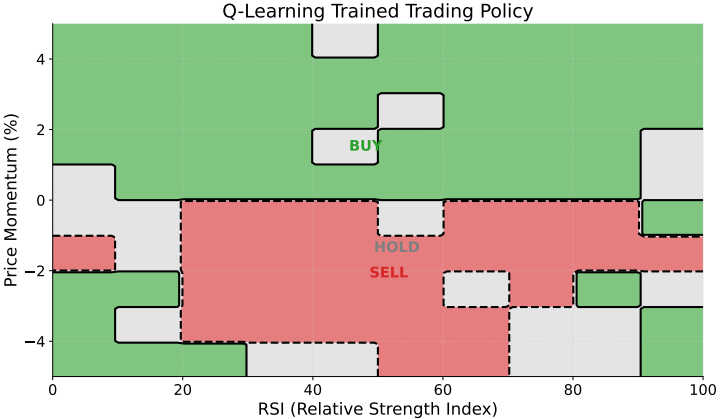
Watching the Agent Learn

- Early episodes: near-random, low reward
- Mid training: discovers good paths
- Late training: converges to optimal policy



The learning curve shows cumulative reward increasing as the agent improves.

Visualizing the Learned Policy



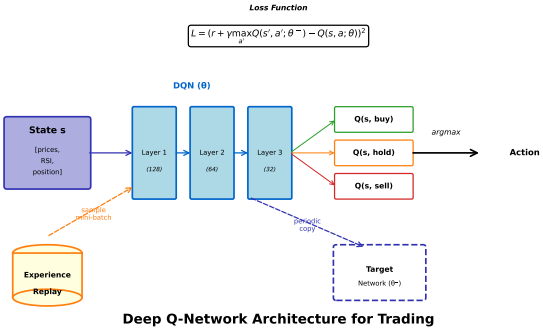
https://github.com/Digital-AI-Finance/methods-algorithms/tree/master/slides/L06_Embeddings_RL/06_policy_viz

Arrows show the optimal action in each state – the agent has learned to navigate.

Deep Q-Networks (DQN)

- Replace Q-table with a **neural network**
- Input: state → Output: $Q(s, a)$ for all a
- **Experience replay**: store and resample transitions
- **Target network**: stabilize training

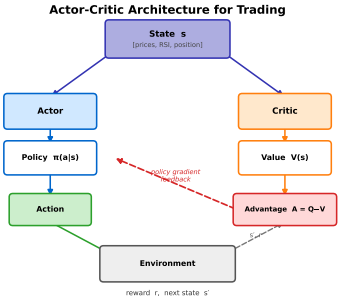
Scales to continuous or very large state spaces.



Deep Q-Network Architecture for Trading

DQN (Mnih et al., 2015) enabled RL to play Atari games from raw pixels.

Why Combine Value and Policy Methods?



Actor-Critic Architecture:

- **Actor**: learns policy $\pi_\theta(a|s)$ directly
- **Critic**: learns value $V_\phi(s)$ to reduce variance
- **Advantage**: $A(s,a) = Q(s,a) - V(s)$
  "Was this action better or worse than average?"

Finance: Actor decides trades, Critic evaluates portfolio value.

Actor-Critic: the foundation for modern algorithms like PPO, A2C, and SAC.

How Does PPO Prevent Catastrophic Updates?

The Clipped Surrogate Objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta)\, A_t,\ \text{clip}(r_t(\theta),\ 1-\epsilon,\ 1+\epsilon)\, A_t)\right]$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the probability ratio.

- If the policy changes too much (r_t far from 1), the objective is **clipped**
- Prevents catastrophic forgetting: no sudden strategy flips
- $\epsilon \approx 0.2$ is the standard clipping range
- PPO is used by OpenAI (ChatGPT via RLHF), DeepMind, and most practitioners

PPO = Proximal Policy Optimization (Schulman et al., 2017). The workhorse of modern RL.

What If You Cannot Explore? Offline RL

Problem: Standard RL requires live interaction. In finance, exploration is costly.

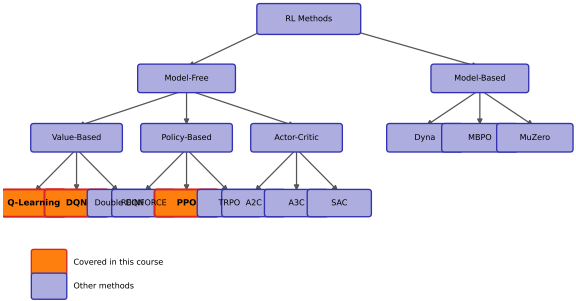
Conservative Q-Learning (CQL):

$$\min_Q \; \alpha \, \mathbb{E}_{s\sim D}\left[\log \sum_a \exp Q(s,a) - \mathbb{E}_{a\sim\hat{\pi}_\beta}[Q(s,a)]\right] + \frac{1}{2}\,\mathbb{E}_{(s,a,r,s')\sim D}\left[\text{TD error}^2\right]$$

- First term: penalize Q-values for unseen actions (conservative)
- Second term: standard Bellman error on the dataset D
- Result: learns a policy that stays close to the data
- **Decision Transformer**: treats RL as sequence modeling (no value function needed)

Offline RL (Levine et al., 2020): learn from historical data without new interaction. Critical for finance.

Where Does Each Method Fit?



RL Method Taxonomy

Actor-Critic: A2C, SAC **Value-based**: Q-learning, DQN **Policy-based**: PPO

Orange = covered in this course. Most practitioners start with PPO (policy-based + Actor-Critic).

Algorithmic Trading as an MDP

MDP Formulation

- **State**: (price, position, time)
- **Actions**: buy, sell, hold
- **Reward**: realized P&L per step

Why RL Fits

- Sequential decisions over time
- Delayed reward (hold for days)
- No "correct label" for each action

The trading agent learns when to enter and exit positions to maximize returns.

Portfolio Optimization with RL

Setup

- **State**: portfolio weights + market features
- **Action**: rebalance weights
- **Reward**: risk-adjusted return (Sharpe ratio)

Advantages

- Handles transaction costs naturally
- Adapts to changing market regimes
- Optimizes multi-period objectives

RL-based portfolio management optimizes the Sharpe ratio while accounting for costs.

Optimal Execution: Minimizing Market Impact

Problem: Execute a large order without moving the market.

1. Split the order into smaller slices over time
2. **State**: remaining shares, time left, market volatility
3. **Action**: how many shares to trade now
4. **Reward**: $-$slippage cost (minimize market impact)

The RL agent learns to trade more aggressively in calm markets and slow down during volatile periods.

Optimal execution is one of the most commercially successful RL applications in finance.

Reward Engineering: The Hardest Part of RL

Bad Reward Design

- Maximize raw return $\rightarrow$ extreme leverage
- Maximize win rate $\rightarrow$ tiny gains, huge losses
- No drawdown penalty $\rightarrow$ reckless risk

Better Reward Design

- Sharpe ratio: return / volatility
- Drawdown penalty: punish large losses
- Risk-adjusted P&L with position limits

"The reward function IS your specification of the problem."

Misaligned rewards produce agents that technically optimize but behave badly.

RL Challenges in Finance

Technical Challenges

- **Non-stationarity**: markets change over time
- **Partial observability**: you cannot see everything
- **Sample efficiency**: live trading data is expensive

Practical Challenges

- **Sim-to-real gap**: backtests $\neq$ live markets
- **Regime changes**: bull $\rightarrow$ bear invalidates policy
- **Regulatory**: explain decisions to compliance

These challenges are why RL in finance remains an active research frontier.

When to Use RL – and When Not To

Pros

- Handles sequential decision problems
- No labeled data required
- Adapts to changing environments

Cons

- Slow training, high sample cost
- Reward design is hard to get right
- Sim-to-real gap in finance
- Difficult to interpret and debug

Use RL when decisions are sequential and delayed – not for one-shot predictions.

RL vs Supervised Learning vs Optimization

	Supervised	**RL**	**Optimization**
Data	Labeled (x, y)	Reward signal	Objective function
Sequential	No	Yes	Sometimes
Adapts	Retrain needed	Online learning	Re-solve
Interpretable	Model-dependent	Low	High
Training time	Fast	Slow	Medium

Each paradigm has its sweet spot – RL shines for sequential, adaptive problems.

Hands-on: Train a Q-Learning Agent

```
import numpy as np
# 5x5 gridworld: start (0,0), goal (4,4)
Q = np.zeros((25, 4))  # 25 states, 4 actions
alpha, gamma, epsilon = 0.1, 0.99, 0.1

for episode in range(500):
    s = 0  # start state
    while s != 24:  # goal state
        if np.random.rand() < epsilon:
            a = np.random.randint(4)
        else:
            a = np.argmax(Q[s])
        s_next, r = step(s, a)  # environment step
        Q[s,a] += alpha * (r + gamma*np.max(Q[s_next]) - Q[s,a])
        s = s_next
```

Full notebook: `L06_rl.ipynb`

Try the notebook to train your own gridworld agent and visualize the policy.

Key Takeaways

Concepts

- MDP formalizes sequential decisions
- Bellman equation decomposes value recursively
- Q-learning is model-free and off-policy
- DQN scales Q-learning to large state spaces

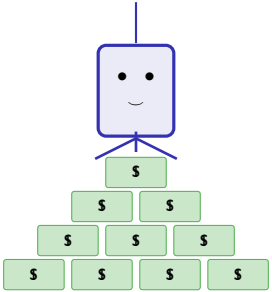
Practice

- Start with tabular Q-learning
- Reward design matters more than algorithm
- Always compare to a simple baseline
- Beware overfitting to backtests

RL is powerful for sequential decisions but demands careful reward engineering.

Closing Thought

"I learned this all by trial and error...
mostly error."



Thank you for a great semester! Good luck with your group assignments.

References

- Sutton, R.S. & Barto, A.G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.
- Watkins, C. & Dayan, P. (1992). Q-learning. *Machine Learning*, 8(3-4), 279–292.
- Mnih, V. et al. (2015). Human-level control through deep reinforcement learning. *Nature*, 518, 529–533.
- Schulman, J. et al. (2017). Proximal policy optimization algorithms. arXiv:1707.06347.
- Levine, S. et al. (2020). Offline reinforcement learning: Tutorial, review, and perspectives. arXiv:2005.01643.
- Chen, L. et al. (2021). Decision Transformer: Reinforcement learning via sequence modeling. *NeurIPS*.
- Liu, X.-Y. et al. (2021). FinRL: Deep reinforcement learning framework for automated stock trading. *SSRN*.

Sutton & Barto: the definitive RL textbook, free at `incompleteideas.net`